Md. Almabud **Juwal**

Senior Software Engineer | B.Sc. Computer Science and Engineering

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Dhaka, Bangladesh

Education

Jan 2020 – Jan 2016 | **B.Sc. in Computer Science and Engineering**

- **Daffodil International University**, Dhaka, Bangladesh (medium of instruction : English)
- **CGPA : 3.77 / 4.00**, 148 credits; result published Jan 2020, degree conferred Jan 2022
- **Relevant coursework (A+) :** Algorithms, Discrete Mathematics, Artificial Intelligence, Compiler Design, Software Engineering, Statistics and Probability, Numerical Methods, Engineering Mathematics

Academic & Research Interests

Artificial Intelligence, Machine Learning, Data Science, Software Engineering, Distributed Systems

Skills

Programming	Python, SQL, JavaScript/TypeScript, Go, C/C++
Frameworks	FastAPI, Django, Django REST Framework, Celery, Angular
Data & Databases	PostgreSQL, Pandas, Redis, MongoDB, Neo4j, Elasticsearch
Machine Learning & AI	scikit-learn, Sentence Transformers, RapidFuzz, Azure Document Intelligence, MCP
Infrastructure & Engineering	Docker, RabbitMQ, Git, CI/CD, Jenkins, Ansible

Experience

Present – Jun 2024 | **Senior Software Engineer | Cogwheel Analytics, Colorado, United States (Remote)**

- Architect the next-generation **Cogwheel Analytics** platform as a **Clean Architecture-based** modular monolith, using reusable application components to reduce repetitive backend implementation
- Lead the development and maintenance of the **Data Cleaning** service, implementing data-validation and transformation workflows to improve data integrity and consistency
- Designed and implemented a hotel-name matching pipeline that combines TF-IDF and fuzzy-string retrieval, sentence-embedding search, and cross-encoder re-ranking with threshold-based acceptance
- Built an MCP server that lets a language model generate custom reports from analytics data, and worked on extracting data from PDF reports with Azure Document Intelligence
- Design and implement backend services, APIs, and asynchronous data-processing workflows using Python, FastAPI, PostgreSQL, and Celery

May 2024 – Jul 2023 | **Senior Software Engineer | Brain Station 23 Ltd, Dhaka, Bangladesh**

- Led the turnaround of a delayed project by mentoring **3 junior engineers**, restructuring task allocation, and redesigning the system architecture, enabling successful delivery of the second version within **6 months**
- Designed software architectures for **Task Station** and **Sayvery**, with a focus on scalability, maintainability, and clear separation of responsibilities
- Managed the technical backlog by translating product and stakeholder requirements into prioritized engineering tasks aligned with project goals and delivery timelines
- Designed and implemented web-scraping and invoice-parsing solutions to automate data collection and extraction

Jun 2023 – Oct 2021 | **Software Engineer | Brain Station 23 Ltd, Dhaka, Bangladesh**

- Improved system performance by approximately **10%** through database-query optimization and API response time by approximately **5%** through application-level caching
- Upgraded Python and Django versions and refactored the existing codebase
- Contributed to real-time synchronization between backend and frontend systems using WebSockets and Go, while mentoring an Associate Software Engineer

Sep 2021 – Jul 2020 | **Junior Software Engineer | Tirzok Private Limited, Dhaka, Bangladesh**

- Reduced processing time for **3.3 million records** from approximately **29–35 minutes to 2–4 minutes** using PostgreSQL COPY and database triggers
- Reduced database query response time by approximately **50%** through query optimization
- Contributed to backend and database architecture, including the implementation of a custom authentication backend and graph-based data solutions using Neo4j
- Automated deployment workflows using Jenkins and Ansible

Apr 2020 – Dec 2019 | **Web Developer | Wizdoor Inc, Dhaka, Bangladesh**

- Contributed to backend and database architecture with frontend and design teams
- Developed a real-time chat application using Django Channels and WebSockets

Projects

Rehive Oct 2021 – Feb 2023

https://rehive.com/

Fintech platform supporting multiple currencies and cryptocurrency. Implemented the billing service and contributed to its extensible plugin architecture.

LEGISLIB May 2020 – Jul 2020

Legal-document processing and search platform that reformatted and indexed case-law PDFs from multiple jurisdictions using MongoDB and Elasticsearch.

Automatic Exam Marks Input (Final-Year Project) 2019

Mobile application that reads student IDs and marks from photographed exam cover pages with the Microsoft Azure Vision API and submits them to the university ERP after review by the faculty member. Team of four, supervised by Dr. S. M. Aminul Haque; responsible for research and development, application code, and the API integration. Grade : A+.

Academic & Programming Achievements

2018 Participant, ACM ICPC Dhaka Preliminary Contest
2018 **4th place**, DIU Intra Team Programming Contest
2017 **2nd place**, DIU Intra Team Programming Contest
2017 **9th place**, DIU Takeoff Individual Programming Contest

Teaching and Mentoring

Dec 2018 – Jan 2018 | **ACM Co-Chair | Computer & Programming Club, Daffodil International University**

- Coordinated programming contests and club activities, and mentored students in competitive programming

Jan 2019 – May 2017 | **Student Prefect | Daffodil International University**

- Supported Competitive Programming and Algorithms lab sessions and assisted the instructor with laboratory examinations and student evaluation

Certifications & Courses

Jun 2025 Advanced Certificate for Management Professionals (ACMP 4.0), University of Dhaka
Apr 2023 AWS Certified Cloud Practitioner
2021 Object-Oriented Design, Coursera (course)
2021 The Ultimate Docker Course, Mosh Hamedani (course)
Sep 2020 HackerRank Python (Basic) Certificate

Languages

Bengali : Native **English** : IELTS Academic 6.5 (CEFR B2) **Hindi** : Conversational